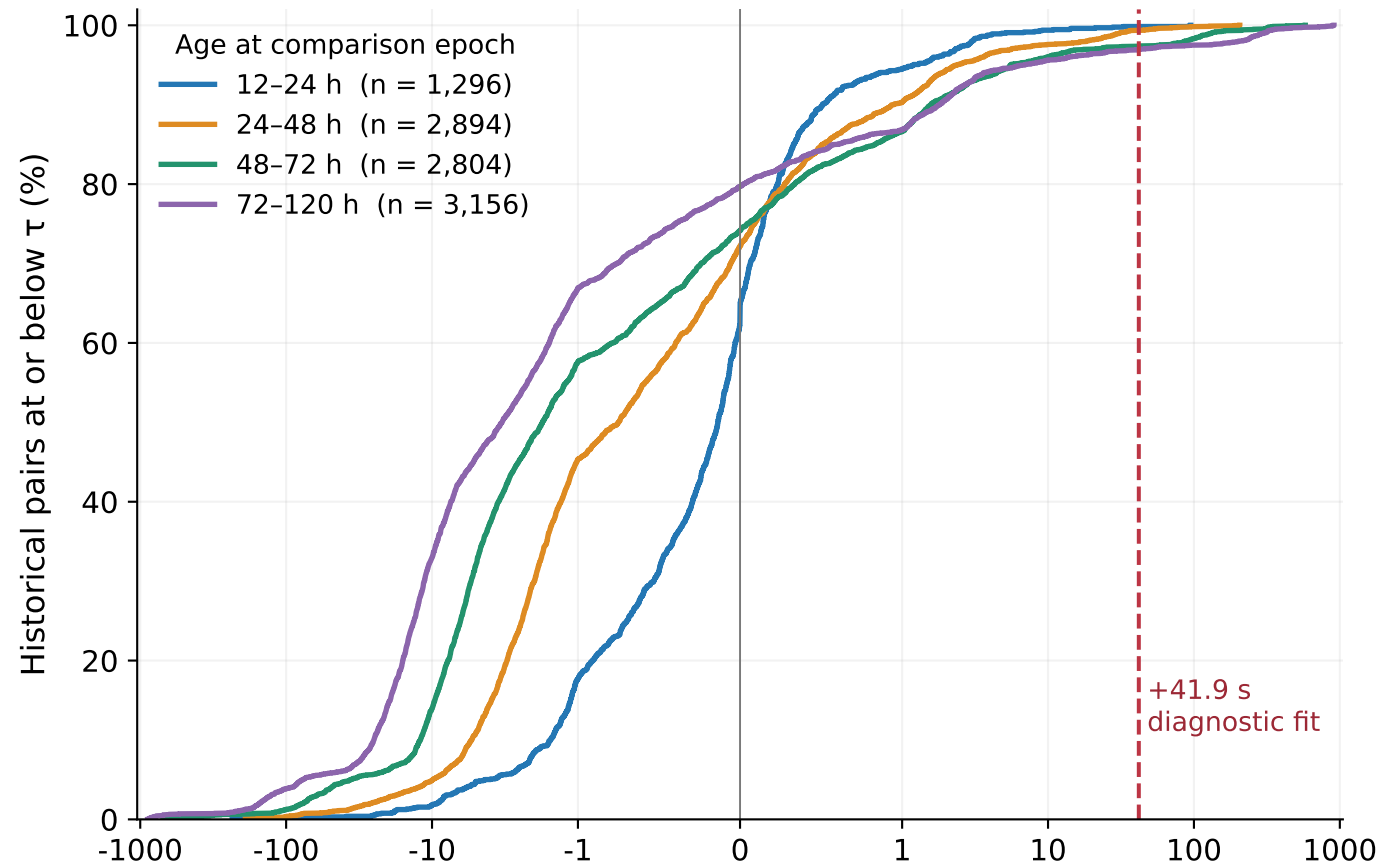


Historical TLE differences expressed as equivalent timing shifts

10,162 element-set pairs · 257 satellites · historical data predating DS5

Observed distributions by older TLE age



Signed equivalent timing difference τ (seconds)

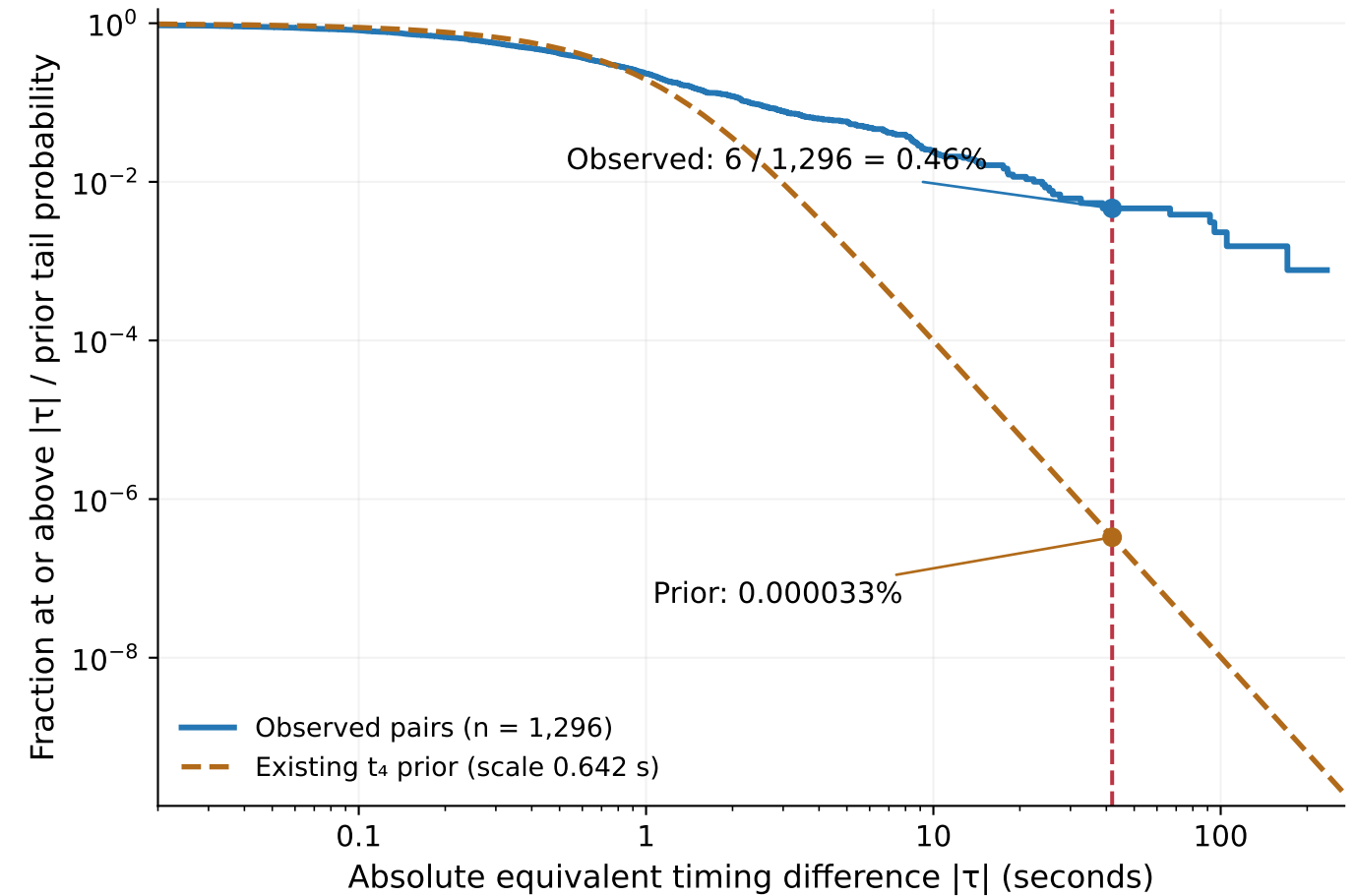
Symmetric-log axis; linear between -1 and $+1$ s

12-24 h observed central 90%: -4.06 to $+1.19$ s. Six pairs exceed ± 41.9 s; only two exceed $+41.9$ s.

These are first-order along-velocity projections of old/new TLE position differences—not verified orbital errors or clock offsets.

Pairs are correlated; each pair has equal weight. No tails trimmed. Under 12 h: only 12 pairs, insufficient to calibrate a young-TLE prior.

12-24 h TLEs: observed tails vs fitted prior



Absolute equivalent timing difference $|\tau|$ (seconds)